

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (CSE) (Credit System) Degree Examinations

January 2023

20CS501 – PROGRAMMING IN JAVA

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1. a)	What is overloading and overriding? Explain. Develop a program demonstrate it.	8	L*3	1	1
b)	Explain with example usage of <i>this</i> and <i>super</i> keywords for variables, constructors and methods.	8	L2	1	1
c)	Differentiate between abstract class and interfaces.	4	L1	1	1
2. a)	Explain the usage of final for variables, method and classes. Demonstrate with the supporting program.	10	L3	1	1
b)	Develop a class called Room with data members length, width and height. Add necessary constructors and method a get Volume() that returns volume of the room. Add necessary code to read dimensions of five rooms from the user and store in an array and print volume of each room.	10	L3	1	1
Unit – II					
3. a)	What is the need of synchronization? Demonstrate with an example program how it is achieved.	8	L3	2	1
b)	What is event handling? Demonstrate how UI events are handled with a simple example.	8	L3	2	1
c)	Explain with a simple example nested try statements.	4	L2	2	1
4. a)	Demonstrate the working of various exception handling constructs with examples.	10	L2	2	1
b)	Develop a program where the main thread reads 10 numbers from the user. Thread A filters the even numbers, thread B filters the odd numbers, thread C merges the result of thread A and B. Main thread prints the result of thread C.	10	L3	2	1
Unit – III					
5. a)	Explain with supporting example various types of swing buttons.	8	L2	3	1
b)	A <i>book.csv</i> file contains book information in the format accession number, title, author, edition, price and publication. Develop a program to read the file and print book title and authors whose price does not exceed ₹500/-.	8	L3	3	1
c)	Highlight any four methods of class File with its description.	4	L1	3	1
6. a)	Develop a swing application with two text fields to read two strings. On click of submit button, the string in the text field should be merged, converted to uppercase and be displayed in a label. Add a reset button to clear the text fields.	10	L3	3	1
b)	Explain JList and JComboBox with necessary code snippets of its usage.	10	L3	3	1
Unit – IV					
7. a)	Develop a JDBC program to modify database using java methods.	8	L3	4	1

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b)	What is the use of InetAddress class? Demonstrate its application.	8	L3	4	1
c)	Briefly explain the types of JDBC Drivers.	4	L2	4	1
8. a)	Explain with necessary coding statements, steps in connecting a java application with a sample database.	10	L2	4	1
b)	Develop a network program to exchange message between client and server using TCP socket.	10	L3	4	1
Unit – V					
9. a)	Explain with example the different types of JSP tags.	10	L2	5	1
b)	Highlight some of the methods of List interface. Demonstrate how collection of mobiles can be managed using a suitable concrete class of List. Develop a simple supporting Mobile class.	10	L3	5	1
10. a)	Develop a java program to illustrate managing Book objects using any suitable class from collection framework. Incorporate the code to display details of books published by a given publisher where the publisher name is entered by the user from keyboard.	10	L3	5	1
b)	Design a html form to read name of the user and a JSP page displaying the name with a greeting message.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
